

Answer to Beal's conjecture

Abstract

The factors of a number are, at least prime numbers (because of the fundamental theorem of arithmetic). To test for common factors of numbers, we can use exclusively prime factors (because they compose others). So for $A=15$, since its composite, has prime factors 3 and 5. If the number is prime, we can only consider one prime factor, since $A=3$, and multiplying several times by A , holds the same prime factors.

Examples:

1. $A = 3, x = 3$

1.1. $A^x = 3 * 3 * 3 = 27$

2. $A = 2*5, x = 3$

2.1 $A^x = 2 * 5 * 2 * 5 * 2 * 5 = 1000$

Also, is only necessary to verify factors of A and B , B and C , or A and C , because if they haven't got a common factor, We don't need to go any further.

So, to ease the problem, i used variables as common factors. For example, we suppose that A^x exists, meaning, for example, that A equals 2 or 3. If $B = A$, then they have common factors.

Trying to prove

If A , and B are prime, they don't share a common factor.

Proof

A prime, and B, prime. If they are different, there is no common prime factor.

$$A^x + B^y = C^z$$

$A \neq B$, and prime, so they have no common factor.

Example:

$2 * 2 * 2 + 3 * 3 * 3 = 35$...dont share common factor.

Given $A=B$, and prime. If C is prime, and equal to A (and B), unless the factor is two, it is not possible

Assuming A is the prime factor.

$$A^x + A^y = A^z$$

For $x = y$ and $z = x$ and $z = y$

$$A^y + A^y = 2(A^y)$$

$$A^z = 2(A^y)$$

$$\frac{A^z}{2} = A^y \tag{1}$$

$$A^z * \frac{1}{2} = A^y \tag{2}$$

$$2*2*2*2 * 1/2 = 2*2*2$$

They all share a factor (factor A), but A^z , must be divisible by 2, meaning, it must have number 2 as factor.

Since $A^x = A^z$, $\frac{A^z}{2} \neq A^x$

Also, $A^z * \frac{1}{2}$, can't be equal to A^y

$$\frac{(2 * 3 * 2)^z}{2} = (2 * 3 * 2)^z.$$

, Not possible. (same number, and z divided by 2)

For $(z > y)$ and $x = y$

From equation 1 we get

$$\frac{A * A * A * A}{2} = A * A * A$$

$$A * A * A * A = 2 * (A * A * A)$$

So conjecture might work if A equals 2. Else no. Because of prime factorization, if A, not equal to 2, number are different. E.g.

$$3 * 3 * 3 * 3 = 2 * 3 * 3 * 3$$

For $z < y$, and $x = y$

From equation 1 we get

$$\frac{A * A * A}{2} = A * A * A * A$$

$$A * A * A * \frac{1}{2} = A * A * A * A$$

Can't be equal

For $y > x$ and $z == y$

$$A^x + A^y = A^z$$

Putting a power in evidence, on left side. we choose A^x , because it has the smaller exponent. (y < x)

$$A^x(1 + A^{y-x}) = A^z \quad (3)$$

divide both sides, by A^x

$$\frac{A^x}{A^x} * (1 + A^{y-x}) = \frac{A^z}{A^x}$$

Since

$$A^x/A^x = 1$$

We get

$$1 + A^{y-x} = \frac{A^z}{A^x} \quad (4)$$

$$1 + A^{y-x} = A^{z-x} \quad (5)$$

Not possible, left side not a power of A's. and $z > x$.

For $y < x$ and $y == z$

Instead of choosing, to put in evidence A^x , we put A^y in evidence

$$A^y(A^{x-y} + 1) = A^z \quad (6)$$

Dividing

$$A^y$$

, by both sides, we get

$$A^{x-y} + 1 = 1$$

...not possible

So methodology, is, if $y \leq x$, put in evidence, A^y , meaning, if we have $A^x + B^y = C^z$, if $x > y$, we put in evidence A^y , else, we put A^x .

In this case we put A^y in evidence

$$A^y(1 + A^{x-y}) = A^z \quad (7)$$

Now we divide A^z by A^y

$$1 + A^{x-y} = A^z/A^y \quad (8)$$

so left side, is either

$$1 + A^{x-y} \text{ or } 1 + A^{y-x} \quad (9)$$

So, we are left with dividing A^z (either use A^x or A^y)

if $z \leq y$, exponent is positive, and this returns A^*A^*A (and so on, z-y times)

$$1 + A^{x-y} = A * A * A * \dots \text{ not possible} \quad (10)$$

something is z-y, because y is bigger

$$1 + A^{x-y} = A^{\text{negative exponent}} \quad (11)$$

$$1 + A^{x-y} = \frac{1}{\text{something}} \quad (12)$$

Doesnt hold.

If we use $1 + A^{y-x}$, is the same

For $y > x$ and $y > z$ and $x > z$
using 5, we get

$$1 + A^{y-x} = A^{-\text{something}}$$

, because $x > z$, exponent is always negative...

So we get, $1 + \text{something (positive)} = \frac{1}{\text{something}}$...doesnt hold.

For $y > x$ and $y > z$ and $x < z$
 using 5, we get

$$1 + A^{y-x} = A^{\text{something}}$$

, because $x \nmid z$, exponent is always positive
 Doesn't hold, left side not on form $x^*x^*x^*x^*\dots$

For $y < x$ and $y > z$ and $x > z$

$$A^x + A^y = A^z$$

$$A^y(1 + A^{x-y}) = A^z$$

$$1 + A^{x-y} = \frac{1}{\text{something}}$$

doesn't hold.

For $y < x$ and $y > z$ and $x < z$

$$A^x + A^y = A^z$$

$$A^y(1 + A^{x-y}) = A^z$$

$$1 + A^{x-y} = A^{x-z}$$

doesn't hold.

For $y > x$ and $y < z$ and $x > z$

using 3

$$1 + (A^{y-x}) = \frac{A^z}{A^y}$$

$1 + (A^{y-x}) =$ division, on right side, is less than one...

Doesn't hold.

For $x > y$

Same as, $y > x$, but switch variables. (All y's are x, and all x are y, on the formulas)

Conclusion

**If A and B, are prime (either equal or not),
they don't share a common factor**

Other remarks

If we want to know the interval between $3 * 3 * 3 + 5 * 5 * 5$, we can get the number 152, factorize it in primes, $2^3 * 19$. So $2 * 2 * 2 < 152 < 2 * 2 * 2 * 19$

(19 is prime, no factorization possible)..so, 152, can't be on form

$$2 * 2 * 2 * 2 \dots \text{or } x * x * x * x \dots$$

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Email : andrealbergaria@gmail.com

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Bibliography

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